Sarang S. Bhagwat

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EDUCATION

Ph.D. 2025 — *University of Illinois Urbana-Champaign*, Urbana, IL, Civil Engineering (Environmental Engineering)
Dissertation: *Linking Process-Level Biorefinery Innovations to Systems-Scale Sustainability*
Advisor: Jeremy S. Guest

M.S. 2019 — *North Carolina State University*, Raleigh, NC, Civil Engineering (Environmental Engineering)
Thesis: *An Environmental Life-Cycle Modeling and Analysis Framework for Microalgal Biofuels*
Advisor: James W. Levis

B.S. 2017 — *Institute of Chemical Technology*, Bombay/Mumbai, India
Chemical Technology (Oils, Oleochemicals, and Surfactants Technology)

SELECTED PROFESSIONAL APPOINTMENTS

2025-present — **Postdoctoral Researcher**
University of California, Berkeley (Energy & Biosciences Institute)
Affiliate, Joint BioEnergy Institute (JBEI), Lawrence Berkeley National Laboratory
Supervisor: Corinne D. Scown

2020-2025 — **Graduate Research Assistant / Student Instructor / Teaching Assistant / Mavis Future Faculty Fellow / Ravindar K. and Kavita Kinra Fellow**
Department of Civil & Environmental Engineering, University of Illinois Urbana-Champaign
DOE Center for Advanced Bioenergy and Bioproducts Innovation (CABBI)

2022-2023 — **Consultant, Process Design & Sustainability Analyses**
Bluestem Biosciences, Inc., Omaha, Nebraska

RESEARCH EMPHASES

Process Systems Engineering; Data Analytics; Circular Economy; Biomanufacturing

Process synthesis, design, and simulation under uncertainty
Discover and prioritize biomanufacturing pathways through physics-informed AI & ML
Develop open-source tools for techno-economic analysis (TEA) and life cycle assessment (LCA)
Navigate economic and environmental trade-offs in R&D decisions

PUBLICATIONS

Peer-Reviewed Journal Articles *(published and in press)*

19. **Bhagwat, S. S.**†; Rao, C.V.; Zhao, H.; Singh, V.; Guest, J. S.† A unifying equation for fermentation sustainability across the titer-rate-yield landscape. *Nature Communications.* 2026, 17 (1), 8655. https://doi.org/10.1038/s41467-026-75285-1. **†Co-corresponding authors.**

18. Lu, J.; Guo, W.; Dong, J.; **Bhagwat, S. S.**; Guest, J. S.; Honaker, R.; Jiao, Y.; Park, D.; Tan, S.; Zhu, Z.; Zhao, H. Bio-based oxalic acid production in *Issatchenkia orientalis* enables sustainable rare earth recovery. *Nature Communications*. 2026, 17 (1), 2193. https://doi.org/10.1038/s41467-026-68957-5.

17. Guo, W.; Kudli, L. P.; **Bhagwat, S. S.**; Bianco, M. F.; Guest, J.S. Co-processing sustainable aviation fuel at petroleum refineries to reduce costs and carbon intensity. *Environmental Science & Technology*. 2026. https://doi.org/10.1021/acs.est.5c17540.

   - **Cover of *Environmental Science & Technology*, February 2026**

16. Tan, S.*; **Bhagwat, S. S.***; Martin, T. A.*; Mishra, S.; Suthers, P. F.; Maranas, C. D.; Singh, V. S.; Guest, J. S.; Zhao, H. High yield production of 3-hydroxypropionic acid using *Issatchenkia orientalis*. *Nature Communications*. 2026, 17 (1), 899. https://doi.org/10.1038/s41467-025-67621-8. ***These authors contributed equally to this work.**

15. Kudli, L. P.; Cortés-Peña, Y. R.; **Bhagwat, S. S.**; Guest, J.S. Characterizing the potential for sustainable azelaic acid production from high-oleic vegetable oil using the two-step oxidative cleavage method. *ACS ES&T Engineering*. 2025. https://doi.org/10.1021/acsestengg.5c00853.
    - **Cover of *ACS ES&T Engineering*, December 2025**

14. **Bhagwat, S. S.**; Dell'Anna, M. N.; Li, Y.; Cao, M.; Brace, E. C.; Bhagwat, S. S.; Huber, G. W.; Zhao, H.; Guest, J. S. Sustainable triacetic acid lactone production from sugarcane by fermentation and crystallization. *ACS Sustainable Chemistry & Engineering*. 2025, 13 (42), 17794-17805. https://doi.org/10.1021/acssuschemeng.5c04797.
    - **Cover of *ACS Sustainable Chemistry & Engineering*, October 2025**

13. Tran, V. G.; Tan, S.-I.; Xu, H.; Weilandt, D. R.; Li, X.; **Bhagwat, S. S.**; Zhu, Z.; Guest, J. S.; Rabinowitz, J. D.; Zhao, H. Decompartmentalization of the yeast mitochondrial metabolism to improve chemical production in Issatchenkia Orientalis. *Nature Communications*. 2025, 16 (1), 7110. https://doi.org/10.1038/s41467-025-62304-w.

12. Kim, M. S.; **Bhagwat, S. S.**; Santiago-Martinez, L.; Shi, X.; Choi, K.; Guest, J. S.; Huber, G. W. Sustainable potassium sorbate production from triacetic acid lactone in food-grade solvents. *Green Chemistry*. 2025, 27 (21), 6087–6104. https://doi.org/10.1039/D4GC04832F.

11. Tran, V. G.*; Mishra, S.*; **Bhagwat, S. S.***; Shafaei, S.; Shen, Y.; Allen, J. L.; Crosly, B. A.; Tan, S.; Fatma, Z.; Rabinowitz, J. D.; Guest, J. S.; Singh, V. S.; Zhao, H. An end-to-end pipeline for succinic acid production at an industrially relevant scale using *Issatchenkia orientalis*. *Nature Communications*. 2023, 14 (1), 6152. https://doi.org/10.1038/s41467-023-41616-9. ***These authors contributed equally to this work.**

10. Yan, Q.; Jacobson, T. B.; Ye, Z.; Cortés-Pena, Y. R.; **Bhagwat, S. S.**; Hubbard, S.; Cordell, W. T.; Oleniczak, R. E.; Gambacorta, F. V.; Vazquez, J. R.; Shusta, E. V.; Amador-Noguez, D.; Guest, J. S.; Pfleger, B. F. Evaluation of 1,2-diacyl-3-acetyl triacylglycerol production in *Yarrowia lipolytica*. *Metabolic Engineering*. 2023, 76, 18–28. https://doi.org/10.1016/j.ymben.2023.01.003.

9. Lee, J. W.; **Bhagwat, S. S.**; Kuanyshev, N.; Cho, Y. B.; Sun, L.; Lee, Y.-G.; Cortés-Peña, Y. R.; Li, Y.; Rao, C. V.; Guest, J. S.; Jin, Y.-S. Rewiring yeast metabolism for producing 2,3-butanediol and two downstream applications: techno-economic analysis and life cycle assessment of methyl ethyl ketone (MEK) and agricultural biostimulant production. *Chemical Engineering Journal*. 2023, 451, 138886. https://doi.org/10.1016/j.cej.2022.138886.

8. Singh, R.*; **Bhagwat, S. S.***; Viswanathan, M. B.*; Cortés-Peña, Y. R.; Eilts, K. K.; McDonough, G.; Cao, M.; Guest, J. S.; Zhao, H.; Singh, V. S. Adsorptive separation and recovery of triacetic acid lactone from fermentation broth. *Biofuels Bioproducts & Biorefining*. 2022, 17, 109-120. https://doi.org/10.1002/bbb.2427. ***These authors contributed equally to this work.**

7. **Bhagwat, S. S.**; Li, Y.; Cortés-Peña, Y. R.; Brace, E. C.; Martin, T. A.; Zhao, H.; Guest, J. S. Sustainable production of acrylic acid via 3-hydroxypropionic acid from lignocellulosic biomass. *ACS Sustainable Chemistry & Engineering*. 2021, 9 (49), 16659–16669. https://doi.org/10.1021/acssuschemeng.1c05441.
   - **Interviewed and quoted in Chemical & Engineering News, C&EN (news article link)**

6. Li, Y.; **Bhagwat, S. S.**; Cortes-Peña, Y. R., Ki, D.; Rao, C. V.; Jin, Y.-S.; Guest, J. S. Sustainable lactic acid production from lignocellulosic biomass. *ACS Sustainable Chemistry & Engineering*. 2021, 9 (3), 1341-1351. https://doi.org/10.1021/acssuschemeng.0c08055.

Peer-Reviewed Journal Articles *(submitted and in review)*

5. Mishra, S.; Chacko, P.K.; Mahata, C.; **Bhagwat, S.S.**; Tran, Vinh G.; Zhao, H.; Guest, J.S.; Singh, V. Evaluation of an industrial-pilot scale succinic acid production and recovery platform using glucose, CO_2(g), and engineered *Issatchenkia orientalis*. *Green Chemistry*. Originally submitted July 21, 2026.
4. Guo, W.; Cortés-Peña, Y. R.; **Bhagwat, S. S.**; Kudli, L. P.; Guest, J. S. Prioritizing early opportunities to enhance carbon efficiency at biorefineries. *Environmental Science & Technology*. Originally submitted July 6, 2026. Pre-print available online: https://doi.org/10.26434/chemrxiv.15005419/v2.
3. Poddar, T. K.; **Bhagwat, S. S.**; Kenny, J.; Aboagye, E.; Shi, R.; Baral, N.R.; Maravelias, C. T.; Field, J.; Guest, J. S.; Scown, C. D. Design considerations and best practices for technoeconomic analysis of new bioprocesses. *Nature Chemical Engineering*. Originally submitted January 15, 2026.
2. Mishra, S.; Augusto, L.; **Bhagwat, S.S.**; Ye, J.; Bari, N.; Nguyen, D.; Zifan, X.; Zhou, Z.; Das, M.; Jia, Y.; Eilts, K.; Mahata, C.; Rao, C.V.; Lu, T.; Jin, Y.S.; Guest, J.S.; Singh, V. A scalable acetate-to-food biomanufacturing platform using synthetic consortia. *Nature Food*. Originally submitted June 20, 2025. Pre-print available online: https://dx.doi.org/10.2139/ssrn.6595420.
1. Dell'Anna, M. N.; **Bhagwat, S. S.**; Dastidar, R. G.; Handy, B. E.; Guest, J. S.; Weller, J.; Dumesic, J. A.; Huber, G. W. Catalytic conversion of triacetic acid lactone to new bio-derived chemicals and fragrances on palladium oxide catalysts. *ACS Sustainable Chemistry & Engineering*. Originally submitted June 10, 2023.

Peer-Reviewed Journal Articles *(in preparation to be submitted by October 2026; only first-author manuscripts included below)*

Bhagwat, S. S.; Sankaranarayanan, K.; Keasling, J.D.; Mukhopadhyay, A.; Scown, C.D. Linking microbial kinetics to facility-scale economic outcomes under uncertainty. Anticipated submittal to [redacted].

Bhagwat, S. S.; Huntington, T.; Chen, Y.; Gong, M.; Scown, C.D. SFF: An AI-Ready Standard Process Flowsheet Format for Interoperability. Anticipated submittal to [redacted].

Bhagwat, S. S.; Kim, J.; Rao, C.V.; Scown, C.D.; Guest, J. S. AutoSynthesis: An agile, generative platform for the conceptual design of sustainable integrated systems. Anticipated submittal to [redacted].

RESEARCH PROPOSALS

8. [redacted] 2026; $300,000; PI: Y. Li; **Senior Personnel & Co-Technical Lead: S. S. Bhagwat.** *Submitted*. (I ideated and co-wrote this proposal with PI Li.)
7. "BioFoundry: NSF iBioFoundry for Basic and Applied Biology"; National Science Foundation (NSF); 2024; $15,000,000; PI: H. Zhao; co-PI: J.S. Guest, others. *<u>Awarded</u>*. (I proposed and co-wrote two technical sections on BioSTEAM software developments.)

6. "Project Bluestem"; Bioindustrial Manufacturing and Design Ecosystem (BioMADE); 2023; $340,740; PI: T. Autera; co-PI: J. Hayes, others. **Key Personnel: S.S. Bhagwat**. *Awarded.* (I performed supporting analyses, set quantitative development targets for fermentative 3-hydroxypropionic acid production, and co-wrote a technical section of this proposal.)
5. "Center for Advanced Bioenergy and Bioproducts Innovation (CABBI): 5-Year Renewal"; US Department of Energy (DOE); 2022–2027; $145,000,000; PI: A. Leakey; Thrust Lead: J.S. Guest, others. *Awarded.* (I was responsible for ideating and co-writing Objective S2.1, one of the three main Sustainability Thrust Objectives.)
4. "New Technologies for Industrial Production of Succinic Acid"; Bioindustrial Manufacturing and Design Ecosystem (BioMADE); 2021–2023; $1,000,000; PI: Huimin Zhao; co-PI: J.S. Guest, others. *Awarded.* (I performed supporting analyses and co-wrote a technical section.)
3. "Novel Modular Treatment System for Distributed Energy Recovery and Water Reclamation from Industrial Wastewaters"; Office of Energy Efficiency and Renewable Energy (EERE), Department of Energy; 2021–2024; $1,628,356; PI: P. Novak; co-PI: J.S. Guest, W. Arnold, P. Grande, J. Griffis, L. Thompson, N. Wright. *Awarded.* (I performed supporting analyses and co-wrote a technical section.)
2. "EFRI E3P: Polyethylene waste as raw material for new, reprocessable thermosets and plastics"; National Science Foundation (NSF); 2021–2024; $1,999,618; PI: D. Guironnet; co-PI: J.S. Guest, C. Evans, K. Smith. *Not funded.* (I performed supporting analyses and co-wrote a technical section.)
1. "Production of Medium-Chain Fatty Acids from Wet Organic Wastes"; Office of Energy Efficiency and Renewable Energy (EERE), Department of Energy; 2020–2023; $2,500,000; PI: A. Delgado; co-PI: J.S. Guest, B. Rittmann, C. Torres. *Not funded.* (I performed supporting analyses and co-wrote a technical section.)

INVITED TALKS *(presenting author underlined)*

9. **Bhagwat, S. S.** Toward agile and predictive process systems engineering for circular economies. Texas Tech University (Department of Chemical Engineering). Lubbock, TX. May 18, 2026.
8. **Bhagwat, S. S.** Toward agile and predictive process systems engineering for circular economies. Arizona State University (School of Sustainable Engineering and the Built Environment). Tempe, AZ. April 3, 2026.
7. **Bhagwat, S. S.**; Zhao, H., Rao, C.V.; Singh, V.; Guest, J. S. (Abstract, Presentation) Generating widely applicable systems-scale insights into fermentation titer-rate-yield. Society for Industrial Microbiology and Biotechnology (SIMB) 47th Symposium on Biomaterials, Fuels, and Chemicals (SBFC). Milwaukee, WI. May 4–7, 2025.
6. **Bhagwat, S. S.;** Guest, J.S.; Rao, C.V.; Zhao, H.; Singh, V. (Abstract, Presentation) Charting development pathways for precision fermentation through agile system analyses. Society for Industrial Microbiology and Biotechnology (SIMB) Annual Meeting. Boston, MA. August 4–7, 2024.
5. **Bhagwat, S. S.;** Guest, J.S.; Rao, C.V.; Zhao, H.; Singh, V. (Educational Seminar, Workshop) Charting development pathways for precision fermentation through agile system analyses. US DOE Bioenergy Technologies Office (BETO). Virtual. June 24, 2024.
4. **Bhagwat, S. S.;** Guest, J.S.; Rao, C.V.; Zhao, H.; Singh, V. (Educational Seminar) Charting development pathways for precision fermentation through agile system analyses. Lawrence Berkeley National Lab's Advanced Biofuels and Bioproducts Process Development Unit (ABPDU) SCIENCE Hour Seminar. Presented virtually to ABPDU and Joint BioEnergy Institute (JBEI) personnel. May 16, 2024.

3. **Bhagwat, S. S.;** Guest, J.S.; Rao, C.V.; Zhao, H.; Singh, V. (Abstract, Presentation) Charting development pathways for precision fermentation through agile system analyses. Society for Industrial Microbiology and Biotechnology (SIMB) 46th Symposium on Biomaterials, Fuels, and Chemicals (SBFC). Alexandria, VA. April 28–May 1, 2024.
2. **Bhagwat, S. S.;** Guest, J.S. (Educational Seminar) Integrating process design, simulation, TEA, and LCA. Integrated Bioprocessing Research Laboratory (IBRL): Fermentation Short Course, Champaign, IL. September 12–14, 2023.
1. **Bhagwat, S. S.;** Dell'Anna, M.N.; Li, Y.; Brace, E.C.; Cortés-Peña, Y.R.; Cao, M.; Huber, G.W.; Zhao, H.; Guest, J.S. (Abstract, Presentation) Sustainable production of sorbic acid via triacetic acid lactone (TAL) from lignocellulosic biomass. ACS Spring 2023 Conference: Crossroads of Chemistry, Indianapolis, IN. March 26–30, 2023.

OTHER CONFERENCE PRESENTATIONS & POSTERS

*(presenting author underlined; * denotes research mentee)*

23. **Bhagwat, S. S.**; Sankaranarayanan, K.; Keasling, J.D.; Mukhopadhyay, A.; Scown, C.D. Linking microbial kinetics to facility-scale economics under uncertainty. American Institute of Chemical Engineers (AIChE) Annual Meeting. Minneapolis, MN. November 8–12, 2026.
22. **Bhagwat, S. S.** (Abstract, Meet-the-Faculty-Candidate Poster) Toward agile and predictive process systems engineering for circular economies. American Institute of Chemical Engineers (AIChE) Annual Meeting. Minneapolis, MN. November 8–12, 2026.
21. **Bhagwat, S. S.**; Sankaranarayanan, K.; Keasling, J.D.; Mukhopadhyay, A.; Scown, C.D. Linking microbial kinetics to facility-scale economics under uncertainty. Society for Industrial Microbiology and Biotechnology (SIMB) 47th Symposium on Biomaterials, Fuels, and Chemicals (SBFC). New Orleans, LA. May 3–6, 2026.
20. **Bhagwat, S. S.** (Abstract, Meet-the-Faculty-Candidate Poster) Linking process-level innovations to systems-scale sustainability. American Institute of Chemical Engineers (AIChE) Annual Meeting. Boston, MA. November 2–6, 2025.
19. **Bhagwat, S. S.;** Zhao, H., Rao, C.V.; Singh, V.; Guest, J. S. (Abstract, Presentation) Generating widely applicable systems-scale insights into fermentation titer-rate-yield. American Institute of Chemical Engineers (AIChE) Annual Meeting. Boston, MA. November 2–6, 2025.
18. **Bhagwat, S. S.;** Kim, J.; Guest, J.S. (Abstract, Presentation) Process engineering and systems analyses at your fingertips using large Language models. Association of Environmental Engineering & Science Professors (AEESP) Research and Education Conference. Duke University, Durham, NC. May 20–25, 2025.
17. Guo, W.*; Kudli, P.L.; **Bhagwat, S. S.**; Vardon, R.D.; Guest, J.S. (Abstract, Poster) Co-processing sustainable aviation fuel at petroleum refineries to reduce costs and life cycle greenhouse gas emissions. AEESP Research and Education Conference; Durham, NC; May 20-22, 2025.
16. Guest, J.S.; **Bhagwat, S. S.**; Bianco, M.F.; Guo, W.; Kudli, L. (Presentation, Abstract) Prioritization of research, development, and deployment for biomass feedstock-to-product pathways. ACS National Meeting & Exposition. San Diego, CA. March 23-27, 2025.
15. **Bhagwat, S. S.;** Guest, J.S.; Rao, C.V.; Zhao, H.; Singh, V. (Abstract, Presentation) Charting development pathways for precision fermentation through agile system analyses. American Institute of Chemical Engineers (AIChE) Annual Meeting. San Diego, CA. October 27–31, 2024.
14. **Bhagwat, S. S.**; Dell'Anna, M. N.; Cao, M.; Zhao, H.; Huber, G. W.; Guest, J. S. (Abstract, Poster) Sustainable Sorbic Acid Production via Triacetic Acid Lactone from Lignocellulosic Hydrolysate

Fermentation Broths. Association of Environmental Engineering & Science Professors (AEESP) Research and Education Conference. Northeastern University, Boston, MA. June 20–23, 2023.

13. **Bhagwat, S. S.**; Dell'Anna, M.N.; Li, Y.; Brace, E.C.; Cortés-Peña, Y.R.; Cao, M.; Huber, G.W.; Zhao, H.; Guest, J.S. (Abstract, Presentation; *Outstanding Podium Presentation Award*) Sustainable Sorbic Acid Production via Triacetic Acid Lactone Crystallization from Fermentation Broths. Environmental Engineering & Science (EES) Symposium, University of Illinois Urbana Champaign. April 14, 2023.
12. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Y.R.; Brace, E.C.; Martin, T.A.; Zhao, H.; Guest, J.S. (Abstract, Poster) Sustainable Production of Acrylic Acid via 3-Hydroxypropionic Acid from Lignocellulosic Biomass. ACS Fall 2022 National Meeting & Exposition: Sustainability in a Changing World, Chicago, IL. August 21–25, 2022.
11. Allen, J.L.*; **Bhagwat, S. S.**; Guest, J.S. (Abstract, Poster) Bio-Based Succinic Acid: A Review of the Technological Drivers of Production Costs and Environmental Impacts. Undergraduate Research Symposium in Chemical and Biomolecular Engineering, hosted by Omega Chi Epsilon, University of Illinois Urbana Champaign. April 1, 2022.
10. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Y.R.; Brace, E.C.; Martin, T.A.; Zhao, H.; Guest, J.S. (Abstract, Presentation; *Outstanding Podium Presentation Award*) Sustainable Production of Acrylic Acid via 3-Hydroxypropionic Acid from Lignocellulosic Biomass. Environmental Engineering & Science (EES) Symposium, University of Illinois Urbana Champaign. April 22, 2022.
9. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Y.R.; Brace, E.C.; Martin, T.A.; Zhao, H.; Guest, J.S. (Abstract, Poster) Sustainable Production of Acrylic Acid via 3-Hydroxypropionic Acid from Lignocellulosic Biomass. Association of Environmental Engineering & Science Professors (AEESP) Research and Education Conference. Washington University at St. Louis. June 28–30, 2022.
8. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Y.R.; Brace, E.C.; Martin, T.A.; Zhao, H.; Guest, J.S. (Abstract, Poster) Sustainable Production of Acrylic Acid via 3-Hydroxypropionic Acid from Lignocellulosic Biomass. DOE Biological Systems Science Division Genomic Science Program - Annual Principal Investigator Meeting. Virtual (due to COVID-19). February 28–March 2, 2022.
7. **Bhagwat, S. S.**; Lee, J.W.; Kuanyshev, N.; Sun, L.; Cho, Y.B.; Lee, Y.; Li, Y.; Cortés-Peña, Y.R.; Guest, J.S.; Jin, Y. (Abstract, Poster) Sustainable Production of Methyl Ethyl Ketone via 2,3-Butanediol from Lignocellulosic Biomass. AIChE Bioenergy Sustainability Conference. Virtual (due to COVID-19). December 9–10, 2021.
6. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Y.R.; Ki, D.; Guest, J.S. (Abstract, Poster) Sustainable Bioproducts from First- and Second- Generation Feedstocks. Center for Advanced Bioenergy and Bioproducts Innovation (CABBI) Annual Science Planning Retreat. Urbana, IL. June 22–23, 2021.
5. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Zhao, H.; Guest, J.S. (Abstract, Presentation) Sustainable Production of Acrylic Acid via 3-Hydroxypropionic Acid from Lignocellulosic Biomass. Environmental Engineering & Science (EES) Symposium, University of Illinois Urbana Champaign. Virtual (due to COVID-19). April 23, 2021.
4. Li, Y.; **Bhagwat, S. S.**; Cortes-Peña, Y.; Ki, D.; Rao, C.; Jin, Y.S.; Guest, J.S. (Abstract, Presentation) Sustainable lactic acid production from lignocellulosic biomass. AIChE Bioenergy Sustainability Conference. Virtual (due to COVID-19). October 13–15, 2020.
3. Li, Y.; **Bhagwat, S. S.**; Cortes-Peña, Y.R.; Ki, D.; Rao, C.V.; Jin, Y.-S.; Guest, J.S. (Abstract, Presentation). Evaluating the Sustainability of Lactic Acid Production from Lignocellulosic Biomass. American Chemical Society (ACS) Fall National Meeting & Exposition. San Francisco, CA. August 17–20, 2020.
2. **Bhagwat, S. S.**; Li, Y.; Cortes-Peña, Y.R.; Ki, D.; Rao, C.V.; Jin, Y.-S.; Guest, J.S. (Abstract, Poster). Separation of Lactic Acid as a High-Value Product of Lignocellulose Fermentation. Center for

Advanced Bioenergy and Bioproducts Innovation (CABBI) Annual Retreat. Urbana, IL. June 22–24, 2020.

1. **Bhagwat, S. S.**; Levis, J.W.; Ranjithan, R.S. (Abstract, Poster). Environmental life-cycle modeling and assessment of microalgae-to-biofuel systems. Environmental, Water Resources, and Coastal Engineering (EWC) Graduate Research Symposium. Raleigh, NC. March 1–2, 2019.

SELECTED AWARDS & HONORS

2026	Outstanding Doctoral Dissertation Award (Honorable Mention), Jacobs Engineering Group, Association of Environmental Engineering & Science Professors (AEESP). Awarded annually to one recipient with one/two honorable mentions.
2023-24	Mavis Future Faculty Fellowship (MF3); Grainger College of Engineering, University of Illinois Urbana-Champaign
2023-24	Ravindar K. and Kavita Kinra Fellowship; Department of Civil & Environmental Engineering, University of Illinois Urbana-Champaign
2023	Chester P. Siess Award for Outstanding Graduate Students in Civil & Environmental Engineering; University of Illinois Urbana-Champaign
2022	Outstanding Podium Presentation Award *(1st place)*, 27th Annual Environmental Engineering & Science Symposium; University of Illinois Urbana-Champaign
2021-present	Tau Beta Pi – The Engineering Honor Society
2021-present	Chi Epsilon – The Civil and Environmental Engineering Honor Society

OPEN-SOURCE SOFTWARE

7. **BioSTEAM** – **The Biorefinery Simulation and Techno-Economic Analysis Modules in Python.**
Core developer
BioSTEAM is an open-source steady-state process simulator in Python to enable biorefinery design, simulation, techno-economic analysis (TEA), and life cycle assessment (LCA) under uncertainty.
Documentation: https://biosteam.readthedocs.io/; Download: https://github.com/BioSTEAMDevelopmentGroup; Manuscript: https://doi.org/10.1021/acssuschemeng.9b07040.

6. **PISCES Standard Flowsheet Format (SFF) – An Interoperable, AI-Ready Format to Represent Chemical Process Flowsheets**
Creator & lead developer
A Standard Flowsheet Format (SFF) for Process Integration & Synthesis using Chemical Engineering Standards (PISCES). SFF is a JSON-based standard to represent chemical process flowsheets for interoperability across process simulators. Currently supports export from BioSTEAM.
Documentation: https://sustainability-software-lab.github.io/pisces-standard-flowsheet-format; Download: https://github.com/sustainability-software-lab/pisces-standard-flowsheet-format;
Manuscript: *In preparation*; Website: https://projectpisces.org/.

5. **NSKinetics – The (Non-)Steady State Kinetics Simulation Package**
Creator & lead developer
NSKinetics is a fast, flexible, and convenient package in Python to simulate steady and non-steady state reaction kinetics and to connect them with techno-economic analysis (TEA) and life cycle assessment (LCA) under uncertainty. NSKinetics enables the construction, simulation, and analysis of reaction systems governed by mass action kinetics or other user-defined rate laws.

Documentation: https://nskinetics.readthedocs.io; Download: https://github.com/sarangbhagwat/nskinetics; Manuscript: *in preparation*.

4. **HXN – Automated Synthesis & Design of Heat Exchanger Networks for Optimized Heat Integration.**
   Creator & lead developer
   HXN is an open-source package in Python that enables the automated synthesis and design of heat exchanger networks to perform optimized heat integration, thus reducing the demand for heating and cooling utilities imposed by a system (and the associated costs and environmental impacts).
   Documentation: https://biosteam.readthedocs.io/en/latest/API/facilities/HeatExchangerNetwork.html#biosteam.facilities.HeatExchangerNetwork;
   Download: https://github.com/BioSTEAMDevelopmentGroup/biosteam/tree/master/biosteam/facilities/hxn

3. **AutoSynthesis – Automated Process Design of Biomanufacturing Chemical Process Flowsheets.**
   Creator & lead developer
   AutoSynthesis is an open-source package in Python to enable automated process synthesis and design of sustainable integrated systems (such as biomanufacturing facilities) through robust superstructure generation and optimization of reaction-separation networks coupled with per-simulation optimization of heat exchanger networks using HXN. A large language model-based interface is currently in development.
   Download: https://github.com/sarangbhagwat/autosynthesis; Manuscript: *In preparation*.

2. **TEAmod – Educational Modules for Techno-Economic Analysis.**
   Creator & lead developer
   TEAmod is an open-source package in Python to facilitate teaching undergraduate and graduate students about techno-economic analysis (TEA) with the objectives of: (i) explaining the concept of TEA and its applications; (ii) describing the steps and input parameters needed to perform a TEA and obtain discounted project cash flows; and (iii) applying TEA to assess financial viability and inform the design of a project or product.
   Documentation: https://teamod.readthedocs.io/en/latest/; Download: https://github.com/sarangbhagwat/teamod.

1. **ContourPlots – Visualization of multi-dimensional data for prioritization of research efforts.**
   Creator & lead developer
   ContourPlots is a toolkit in Python for generating multi-dimensional plots easily, usefully, and legibly.
   Download: https://github.com/sarangbhagwat/contourplots

TEACHING & MENTORING

Graduate Student Instructor & ***Teaching Assistant***

Course: CEE 493: Sustainable Design of Engineered Technologies
Department of Civil & Environmental Engineering
University of Illinois Urbana-Champaign
Fall 2023 (Instructor: Jeremy Guest)
Fall 2022 (Instructor: Jeremy Guest)
Contributions:
[Fall 2023] Delivered 4 lectures, taught life cycle costing and techno-economic analysis, taught uncertainty and sensitivity analyses, and taught use of Python and MS Excel VBA for each of those analyses. Developed slides, open-source software, and other course materials for each lecture.
[Fall 2022] Delivered 2 lectures, held office hours, provided feedback on design projects, taught uncertainty and sensitivity analyses, taught coding in Python and MS Excel VBA, and graded for 55 students.

Research Mentor

<u>Undergraduate Students</u> *([duration] name; program; year graduated; placement)*
[May 2021–May 2022] Jayne Allen; BS in Chemical & Biomolecular Engineering (hired as a mentee through DOE CABBI's Research Internship in Sustainable Bioengineering, RISE); 2022; joined the J.S. Guest research group at UIUC as an MS student in Fall 2022.

<u>Graduate Students</u> *([duration] name; program; year graduated; placement)*
[August 2024–present] Princy Chacko; PhD in Chemical & Biomolecular Engineering; 2029 (anticipated); current J.S. Guest research group member.
[August 2024–present] Kaveri Srivastava; PhD in Chemical & Biomolecular Engineering; 2029 (anticipated); current J.S. Guest research group member.
[August 2023–present] Wenjun Guo; PhD in Civil & Environmental Engineering; 2028 (anticipated); current J.S. Guest research group member.
[August 2023–present] Lavanya Kudli; PhD in Civil & Environmental Engineering; 2027 (anticipated); current J.S. Guest research group member.
[August 2021–July 2023] Lavanya Kudli; MS in Civil & Environmental Engineering; 2023; continuing in J.S. Guest research group as a PhD student.
[June 2022–May 2024] Jayne Allen; MS in Civil & Environmental Engineering; 2024; Process Engineer, Archer-Daniels-Midland.

PROFESSIONAL & UNIVERSITY SERVICE & ACTIVITIES

Conference Organization

Session Co-chair: Grand Challenges in Chemical Engineering for a Sustainable Future. Sustainable Engineering Forum (SEF), American Institute of Chemical Engineers (AIChE) Annual Meeting. Boston, MA. November 2–6, 2025.

Session Co-chair: Early-career researchers in sustainable energy. Sustainable Engineering Forum (SEF), American Institute of Chemical Engineers (AIChE) Annual Meeting. Boston, MA. November 2–6, 2025.

Member/Volunteer: Organizing subcommittee for the annual Women Empowered in STEM (weSTEM) conference; GradSWE (Graduate Society of Women Engineers), University of Illinois Urbana-Champaign. 2023-24.

Public Relations Co-organizer: Environmental, Water Resources, and Coastal Engineering (EWC) Graduate Research Symposium. North Carolina State Univ., Raleigh, NC. Feb–March 2019.

Journal Peer Review & Other Service

Editorial Advisory Board: GCB Bioenergy. June 2026 – present.

<u>Peer Reviews</u>: 13
Environmental Science & Technology (2025)
Green Chemistry (2026)
Bioresource Technology (2023)
ACS Sustainable Chemistry & Engineering (2025, 2026)
ACS Industrial & Engineering Chemistry Research (2025, 2025)
RSC Sustainability (2025)
Journal of Industrial Ecology (2026, 2026)
GCB Bioenergy (2026)
Discover Applied Sciences (2025)
Journal of Environmental Chemical Engineering (2026)

University Service

Committee Member: Science Leadership and Management (SLAM) Workshop Series. California Institute for Quantitative Biosciences (QB3), University of California, Berkeley. Fall 2025–Spring 2026.

Speaker: Workshop on writing and presenting preliminary PhD proposals. American Association for Aerosol Research: UIUC Student Chapter, University of Illinois Urbana-Champaign. May 18, 2023.

Judge: Annual Undergraduate Research Symposium. University of Illinois Urbana-Champaign. April 27, 2023.

Volunteer: Vortex–the Chemfest. Institute of Chemical Technology, Mumbai, MH, India. 2015.

Outreach Activities

Panelist & Speaker: Mixed Faculty-Graduate Student Panel, Engineering Open House (annual outreach event). Grainger College of Engineering, University of Illinois Urbana-Champaign. April 5, 2024.

Presenter: Research Fair, Engineering Open House (annual outreach event). Grainger College of Engineering, University of Illinois Urbana-Champaign. April 5, 2024.

External News Media Mentions

Interview & Press Article: "Firms plan pilot plants for biobased acrylic acid". *Chemical & Engineering News* Volume 101, Issue 35. October 17, 2023. Interviewed & quoted for acrylic acid TEA-LCA expertise. https://cen.acs.org/business/biobased-chemicals/Firms-plan-pilot-plants-biobased/101/i35

Community Leadership & Service

Founder & Vice President (2015-16); Secretary, Literary Avenue (2016-17); Member (2015-2017): Rotaract Club of King's Circle, Matunga. Affiliated events with the Institute of Chemical Technology, Mumbai, MH, India. 2015–17.

CONTACT REFERENCES

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